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Experience

•Rakuten Mobile Inc. *Jan. 2024 – Present*

Data Scientist Tokyo, Japan

- **Telco-domain SLM:** Built the RAN-domain data pipeline and evaluation harness for a Qwen3-1.7B adapted via CPT, SFT, and DPO for edge deployment; designed a 75/25 telco/generic CPT mix to curb catastrophic forgetting.
- The harness scored models with a GPT-4o judge and BERTScore over **∼7K Q&A pairs** across 7 telco domains; benchmarking chose the 1.7B over a 4B candidate at **matched quality and ∼2.5× lower CPU latency**.
- **Autocall optimization:** Built a scoring system that picks the best hour to call churn-risk and payment-default customers from 2-week usage history, consistently adding **+23–30M JPY/month** in collections.
- **Agentic analytics:** Built a multi-agent LLM system (Pydantic AI, MinIO data lake + ClickHouse, served via Viber) for natural-language data Q&A, scheduled HTML/PPT reports on churn, onboarding and app usage, plus daily web-scraped competitor-campaign monitoring — saving **∼40 analyst-hours/month**.
- **AI marketing content generation tool:** Designed and built, end to end, a web app that turns a campaign brief into branded HTML emails and SMS (GPT for copy, Gemini for images), grounded in 200+ past SFMC campaigns and their DOMO open/click rates, with 5 audience-targeted variants per request and SVG export for editing in Figma; used by a team of **5 marketers**, cutting draft time from **15 to 5 hours**.

Selected Projects

•TinyNNUE Edge – int8 Inference *2026*

Quantization + integer C++ inference (personal) GitHub · HF · Writeup

- Trained a small chess evaluation network on 121K Stockfish-labeled positions and quantized it to int8 using integer-only math. Same accuracy as fp32 (**147.1 vs 146.9** cp error) at **4**× smaller size.
- Wrote the inference engine in C++, updating only the pieces a move touches instead of recomputing the first layer, with results identical to a full recompute. On CPU, int8's **7.2**× speedup came from unvectorized fp32; vectorized fp32 was **2.5**× faster.

•Ukiyo-e Haiku VLM *2026*

Bilingual vision-language model (personal) GitHub · HF · Demo

- Bilingual (JP/EN) multimodal generation model producing haiku from ukiyo-e prints; curated paired image–text data and fine-tuned via SFT then preference optimization (ORPO / KTO) with a λ_U sweep.
- Shipped two models to the Hugging Face Hub with a live interactive demo (flagship JP model 4.36/5 avg eval), showing preference gains require a real chosen/rejected quality gap.

•Math SLM (SFT + DPO) *2026*

LLM post-training pipeline (personal) GitHub · HF

- SFT + DPO on DeepSeek-R1-Distill-Qwen-7B; teacher self-distillation data generation via vLLM (TP=4), LoRA training with DeepSpeed ZeRO-3 on 8×H100.
- Built the full pipeline (data gen → SFT → LoRA merge → eval); SFT added **+6.4 pp** average on math benchmarks; model live on the HF Hub.

Technical Skills

- **Languages**: Python, C/C++, SQL
- **ML / LLMs**: Post-training quantization (int8, fixed-point integer inference), LLM & VLM fine-tuning + multimodal generation (SFT, DPO, ORPO, KTO), knowledge distillation, CPT, LoRA / PEFT, evals (LLM-as-judge, BERTScore), churn & time-series forecasting
- **Frameworks**: PyTorch, Hugging Face (Transformers, PEFT, TRL), vLLM, Apache TVM (basic), DeepSpeed (ZeRO-3), NumPy, Pandas, Matplotlib
- **Infrastructure**: Multi-GPU training (A100 / H100), distributed parallelism (TP / DP / ZeRO), Git

Education & Achievements

•IIT Indore, B.Tech, Mechanical Engineering (Minor: HSS) — CGPA 8.41 *2019 – 2023*

•JEE Advanced — All India Rank 4879, among 245,000 qualified candidates *2019*